

Unit 11

PARTICLE PHYSICS

53 items: 23 from official papers (153 marks), 30 from the booklet

Syllabus sections in this unit

11.1 Atoms, nuclei and radiation

11.2 Fundamental particles

Unit 11: Particle physics

1. M/J 26/P22/Q6/a,c,d

- 6 (a) One of the conclusions from the α -particle scattering experiment was that the atom is mostly empty space.

State the observation from the experiment that provides evidence for this conclusion.

[1]

- (c) Determine the number of up quarks in a nucleus of ${}^2_1\text{H}$.

number =

[1]

- (d) Describe the general quark composition of a meson.

[1]

2. M/J 26/P23/Q5/a,b,c

- 5 (a) State what is meant by a meson.

[1]

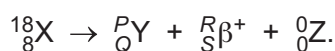
- (b) The elementary charge is e .

Explain, in terms of quark composition, how a baryon may have an overall charge of:

- $-1e$
- $+2e$

[3]

- (c) The nuclide ${}^{18}_8\text{X}$ undergoes β^+ decay to produce nuclide ${}^P_Q\text{Y}$ and one other particle Z, as shown in the equation



- (i) State the values of P , Q , R and S .

$P =$

$Q =$

$R =$

$S =$

[2]

- (ii) State the name of particle Z.

[1]

- (iii) State the net change in quark composition as nuclide X turns into nuclide Y.

[1]

- (iv) Explain why the emission of particle Z results in the β^+ particles from the decays of different nuclei of ${}^{18}_8\text{X}$ having a continuous range of kinetic energies.

[2]

3. M/J 26/P24/Q7/a,b

- 7 (a) (i) State what is meant by isotopes.

[2]

- (ii) A thorium-232 nucleus ($^{232}_{90}\text{Th}$) undergoes radioactive decay to form another nucleus. A sequence of further decays occurs until a nucleus is formed that is another isotope of thorium. In each decay, either a single α -particle or a single β^- particle is emitted.

Determine the minimum numbers of α -particles and β^- particles that could be emitted during the sequence of decays to form the other isotope of thorium from thorium-232.

minimum number of α -particles =

minimum number of β^- particles =

[2]

- (iii) Complete Figure 7.1 to show the nucleon number and proton number of the thorium isotope formed by the sequence of decays.

Th

Figure 7.1

[2]

- (b) (i) State the quark composition of a proton.

[1]

- (ii) Describe the change to the quark composition of the proton when it undergoes β^+ decay.

[1]

4. MAR 26/P22/Q7/a,b,c

- 7 (a) Describe the structure of a nuclear atom.

[2]

- (b) P and Q are radioactive nuclei. P has 94 protons. Q has 91 protons.

Each nucleus decays to form a nucleus with 92 protons.

- (i) State the type of radioactive decay experienced by P.

[1]

- (ii) State the type of radioactive decay experienced by Q.

[1]

- (c) Particle X has a ratio of $\frac{\text{charge}}{\text{mass}}$ equal to $+1.76 \times 10^{11} \text{ C kg}^{-1}$.

Identify particle X. Show your working.

particle X

[2]

5. O/N 25/P21/Q6/a,b,c

6 The nuclide ${}^3_1\text{H}$ is an isotope of hydrogen that is called tritium.

(a) (i) Determine the numbers of protons, neutrons and electrons in a neutral atom of tritium.

number of protons =

number of neutrons =

number of electrons =

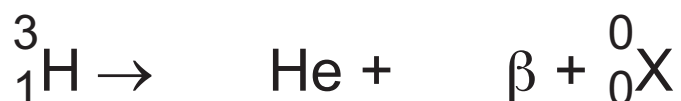
[2]

(ii) Draw a labelled diagram to represent a simple model of the arrangement of the protons, neutrons and electrons in a tritium atom.

[2]

(b) Tritium is radioactive and undergoes β^- decay to form an isotope of helium (He). Gamma radiation is not emitted during this decay.

(i) Complete the equation to represent the radioactive decay of tritium.



[2]

(ii) State the name of particle X.

[1]

(c) Determine the quark composition of a tritium nucleus.

[2]

6. O/N 25/P22/Q6/a,b,c

- 6 (a)** State what is meant by a fundamental particle.

[1]

- (b) (i)** Particle Q is a meson with a charge of 0.

Determine a possible quark composition for Q.

[2]

- (ii)** Particle Q has a mass of 0.67 u and a kinetic energy of 2.1×10^{-16} J.

Calculate the speed of particle Q.

speed = ms^{-1} [3]

- (c)** Radium-228 ($^{228}_{88}\text{Ra}$) is a radioactive nuclide.

- (i)** State the number of electrons in a neutral atom of radium-228.

number of electrons = [1]

- (ii)** A nucleus of radium-228 undergoes a series of decays to form nucleus X. During the process, 5 α -particles and 4 β^- particles are emitted.

Determine the number of protons and the number of neutrons in nucleus X.

number of protons =

number of neutrons =

[2]

7. O/N 25/P24/Q6/a,b,c

- 6** Fig. 6.1 shows four alpha particles W, X, Y and Z moving towards a gold nucleus that is in thin gold foil.

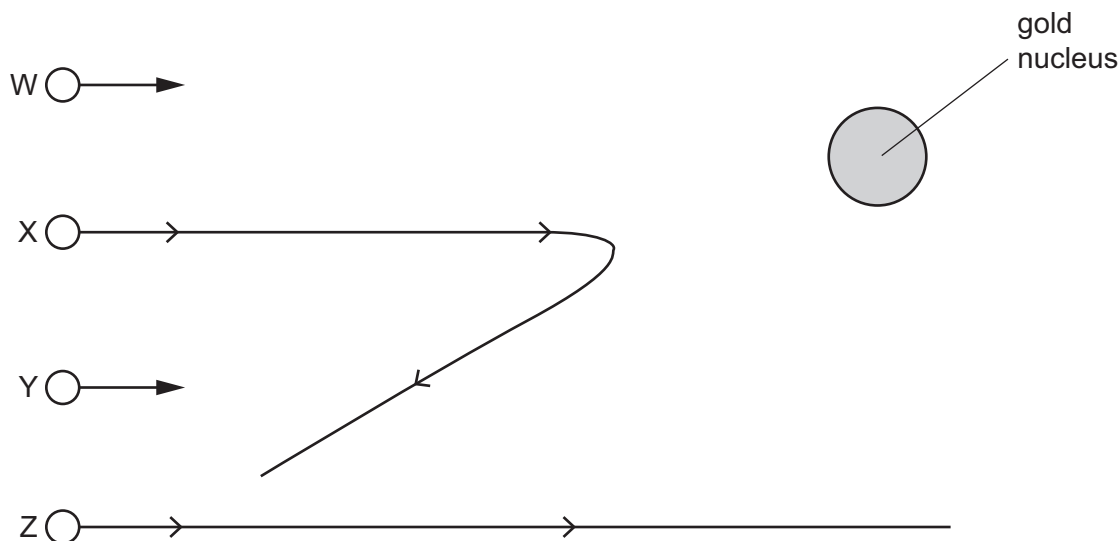


Fig. 6.1 (not to scale)

- (a) (i)** The paths of particles X and Z are shown.
Complete Fig. 6.1 to show possible paths for particles W and Y. [3]
- (ii)** Describe what may be inferred about the structure of an atom from the path of particle X. [1]
- (iii)** When a beam containing many alpha particles is incident on thin gold foil, nearly all of the alpha particles follow paths that are similar to the path of particle Z.
Describe what may be inferred from this about the structure of an atom. [1]
- (b)** State the mass and the charge, in terms of the atomic mass unit u and the elementary charge e , of an alpha particle.
- | | |
|----------|-----|
| mass = | u |
| charge = | e |
- [2]
- (c)** There are two types of hadron.
The hadrons that are in alpha particles are each composed of three quarks.
- (i)** State the name of this type of hadron. [1]
- (ii)** Show, by reference to their constituent quarks, that the hadrons in an alpha particle have charges of either zero or $+1e$. [2]

8. M/J 25/P21/Q7/a,b

- 7 (a) Nitrogen-12 ($^{12}_7\text{N}$) is an unstable isotope of nitrogen that decays by the emission of radiation to carbon-12 ($^{12}_6\text{C}$).

Complete the full nuclear equation for the decay, including all the particles involved.



[3]

- (b) Hydrogen-1 (^1_1H) is an isotope of hydrogen. An atom of hydrogen-1 comprises a proton with an orbiting electron.

An antiparticle equivalent of hydrogen-1 comprises an antiproton with an orbiting positron. The antiquarks in the antiproton are the antiparticles of the quarks in a proton.

- (i) State the charge on the positron in terms of the elementary charge e .

charge = e [1]

- (ii) State the group (class) of fundamental particle to which the positron belongs.

[1]

- (iii) In Table 7.1, state the flavour and charge of the three antiquarks that comprise the antiproton.

Table 7.1

flavour	charge / e

[3]

9. M/J 25/P22/Q7/a,b,c

- 7 (a) State what is meant by a fundamental particle.

[1]

- (b) A nucleus X has 14 nucleons and p protons. The ratio of charge to mass for nucleus X is $4.1 \times 10^7 \text{ C kg}^{-1}$.

- (i) Determine p .

$$p =$$

[3]

- (ii) Nucleus X undergoes β^- decay to form nucleus Z.

Complete the equation representing this decay.



[3]

- (c) A sample of a radioactive substance emits particles that are positively charged and have a continuous range of kinetic energies.

State and explain whether the nuclei in the sample are undergoing α -decay, β^+ decay or β^- decay.

[2]

10. M/J 25/P23/Q8/a,b

- 8 (a) An antiparticle equivalent of the neutron is called the antineutron. The quarks in the antineutron are the antiparticles of the quarks in a neutron.

The elementary charge is e .

In Table 8.1, state the flavour and charge of the three antiquarks that comprise the antineutron.

Table 8.1

flavour	charge / e

[3]

- (b) In β^- decay, a neutron decays to form a proton.

Theory predicts that an antineutron should decay to form an antiproton. A particle and an antiparticle should also be observed.

Suggest the names of the particle and the antiparticle.

particle:

antiparticle:

[2]

11. M/J 25/P24/Q7/a,b,c

- 7 (a) State the names of **two** different leptons.

1

2

[2]

- (b) In the following list, underline all the particles that are hadrons.

antineutrino beta-plus meson neutron

[1]

- (c) By reference to quark composition, show that the charge of a proton is $+1.6 \times 10^{-19} \text{C}$.

[2]

12. MAR 25/P22/Q7/a,c

- 7 An isolated stationary nucleus Q decays into nucleus R and an α -particle. The α -particle has speed $1.5 \times 10^7 \text{ms}^{-1}$.

- (a) Complete the equation for this decay.



[1]

- (c) State **three** quantities that are conserved during the decay.

1

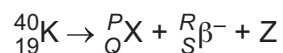
2

3

[3]

13. O/N 24/P21/Q5/a,b

- 5 (a)** Potassium-40 (${}^{40}_{19}\text{K}$) undergoes β^- decay to form a nuclide of element X. Particle Z is emitted during the decay. The equation for the decay is shown.



- (i)** State the values of P , Q , R and S .

$P =$

$R =$

$Q =$

$S =$

[2]

- (ii)** State the name of particle Z.

[1]

- (iii)** State the name of the class of fundamental particle to which both the β^- particle and particle Z belong.

[1]

- (b)** Determine the quark composition of an alpha-particle.

quark composition

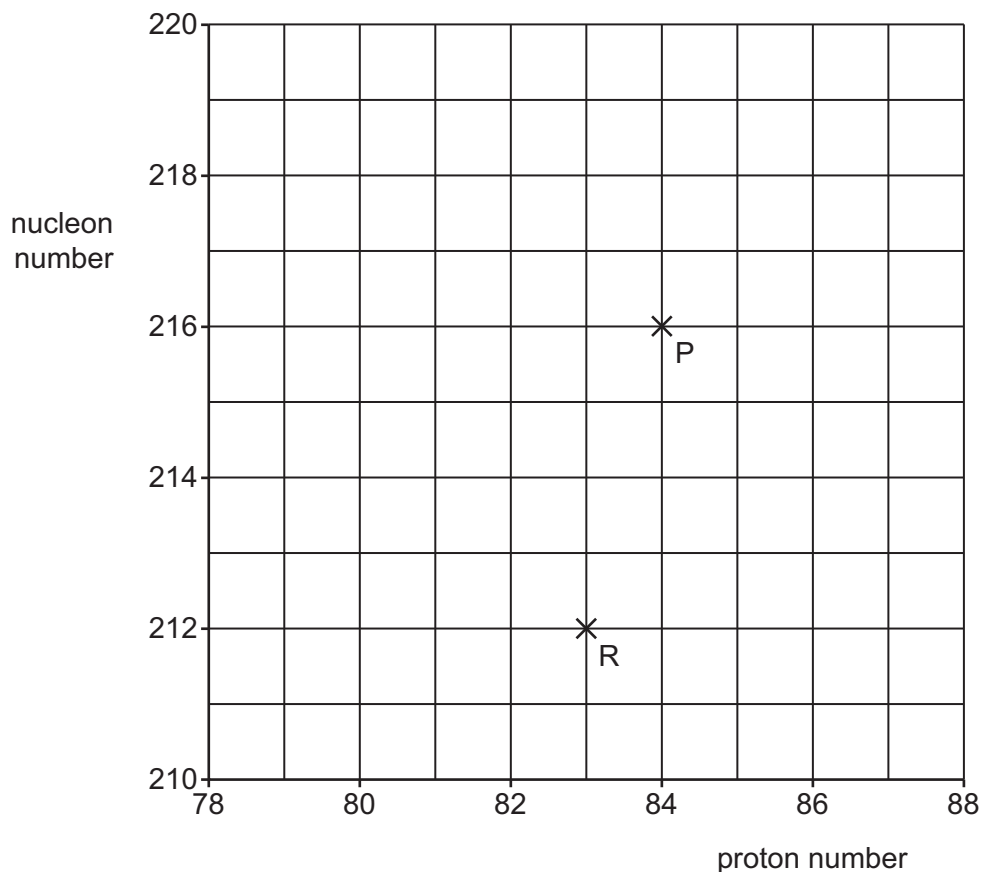
[3]

14. O/N 24/P22/Q6/a,b

- 6** (a) Compare an α -particle with a β^+ particle in terms of their masses and charges.

[3]

- (b) Nucleus P undergoes α -decay to form nucleus Q. Nucleus Q then undergoes a further decay to form nucleus R. The proton and nucleon numbers of P and R are shown in Fig. 6.1.

**Fig. 6.1**

- (i) On Fig. 6.1, draw a cross ($\times$) to show the proton number and nucleon number of Q. Label your cross Q. [1]
- (ii) State the names of the particles emitted as Q decays to form R. [2]

15. O/N 24/P23/Q7/a,b,c

- 7 (a)** Complete Table 7.1 to show the charges, in terms of the elementary charge e , on each of the flavours of quark and antiquark shown.

Table 7.1

flavour	charge / e	
	quark	antiquark
up		
down		
strange		

[3]

- (b) (i)** State the name of the class (group) of fundamental particles to which baryons and mesons belong.

[1]

- (ii)** Compare baryons and mesons in terms of their constituent particles.

[2]

- (c)** Describe β^+ decay in terms of the fundamental particles involved.

[2]

16. M/J 24/P21/Q7/a,b

- 7** Nuclei of an isotope of copper (Cu) each have 29 protons and 37 neutrons. This isotope is a β^- emitter.

(a) State the nuclide notation in the form ${}^A_Z\text{X}$ for this nucleus of copper.

[1]

- (b)** The energy spectrum of the β^- radiation emitted by a sample of this isotope is shown in Fig. 7.1.

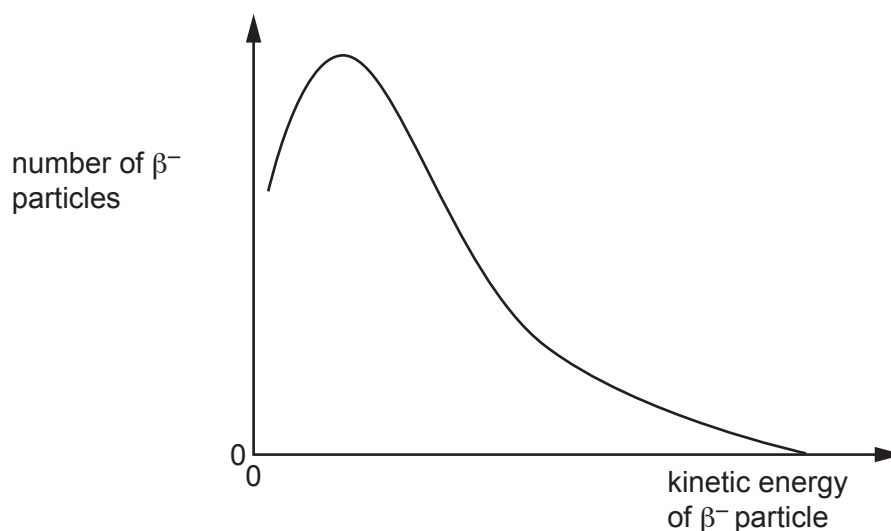


Fig. 7.1

- (i)** Use Fig. 7.1 to explain why other particles apart from the β^- particles must be emitted during this decay.

[3]

- (ii)** State the name of the other particle emitted during the decay of this isotope.

[1]

- (iii)** The copper isotope decays to an isotope of zinc (Zn).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

[3]

17. M/J 24/P22/Q7/a,b

7 A particle Q and a particle R are each composed of one quark and one antiquark.

(a) State the name of the class (group) of particles that includes Q and R.

[1]

(b) Q has a charge of $-1e$, where e is the elementary charge. R has a charge of 0.

Complete Table 7.1 to show a possible second quark in each of Q and R.

Table 7.1

	charge	first quark	second quark
Q	$-1e$	strange	
R	0	anti-up	

[2]

18. M/J 24/P23/Q6/a,b,c

6 Nuclei of an isotope of samarium (Sm) each contain 62 protons and 85 neutrons.

(a) State the nuclide notation in the form ${}^A_Z\text{X}$ for this isotope of samarium.

[1]

(b) This isotope of samarium is radioactive and decays by emitting particles. Gamma-radiation is **not** emitted. The energy spectrum of the emitted particles is shown in Fig. 6.1.

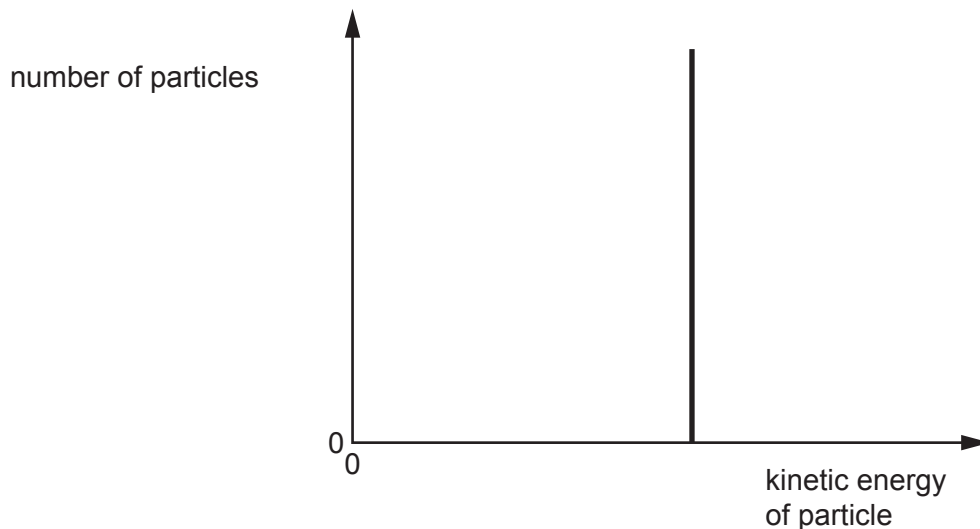


Fig. 6.1

(i) Explain how Fig. 6.1 shows that this isotope of samarium emits α -particles and does **not** emit β -particles.

[2]

(ii) This isotope of samarium decays to an isotope of neodymium (Nd).

Give the radioactive decay equation for this decay. Include the nucleon and proton numbers of **all** the particles involved.

[2]

(c) A baryon is composed of three quarks which all have different flavours. The baryon has a charge of 0.

Two of the quarks in the baryon are an up quark and a bottom quark.

(i) Determine, in terms of the elementary charge e , the charge on the third quark in the baryon.

charge = e [2]

(ii) State a possible flavour for the third quark in the baryon.

[1]

19. MAR 24/P22/Q4/a

4 A nucleus P undergoes α -decay to form nucleus Q.

(a) Complete the equation for this decay.

$${}_{84}^{215}\text{P} \rightarrow \underline{\hspace{1cm}}\text{Q} + \underline{\hspace{1cm}}\alpha \quad [2]$$

20. MAR 24/P22/Q8/a,b,c

8 (a) State the name of the class (group) of fundamental particles that contains a neutrino.

[1]

(b) A hadron P has a charge of $+1e$, where e is the elementary charge. The hadron P is composed of a down antiquark and only one other quark.

(i) Identify a possible flavour for this other quark.

[1]

(ii) State what type of hadron is P.

[1]

(c) Nucleus Q undergoes radioactive decay to form nucleus R, emitting an antineutrino and another particle X, as shown in the decay equation.

$$\text{Q} \rightarrow \text{R} + \text{X} + \bar{\nu}$$

(i) State what particle is represented by X.

[1]

(ii) Compare the nucleon numbers of Q and R.

[1]

(iii) Compare the charges of Q and R.

[1]

21. O/N 23/P21/Q7/a,b,c,d

- 7 (a)** The results of the α -particle scattering experiment led to the development of the nuclear model for the atom.

State the results that suggested that most of the mass of the atom is concentrated in a very small region and most of the atom is empty space.

[2]

- (b)** State the composition of γ -radiation.

[1]

- (c)** Table 7.1 lists the names of three particles and possible classifications for them.

Table 7.1

particle name	classification		
	baryon	hadron	lepton
neutrino			
neutron			
positron			

Complete Table 7.1 by placing ticks (✓) in the boxes to indicate the classifications that apply to each particle.

[2]

- (d)** The discovery of a particle with an unusual charge was an important step in the development of the theory of quarks. The particle is a hadron with a mass of 2.19×10^{-27} kg and a charge of $+2e$, where e is the elementary charge.

- (i)** Calculate the mass, in u , of the particle. Give your answer to three significant figures.

mass = u [1]

- (ii)** Determine a possible quark composition of a hadron with a charge of $+2e$. Explain your reasoning.

[2]

22. O/N 23/P22/Q7/a,b,c

- 7 (a)** In the following list, underline **all** the particles that are **not** fundamental.

antineutrino baryon nucleon positron [1]

- (b)** A nucleus of thorium-230 ($^{230}_{90}\text{Th}$) decays in stages, by emitting α -particles and β^- particles, to form a nucleus of lead-206 ($^{206}_{82}\text{Pb}$).

Determine the total number of α -particles and the total number of β^- particles that are emitted during the sequence of decays that form the nucleus of lead-206 from the nucleus of thorium-230.

number of α -particles =

number of β^- particles =

[2]

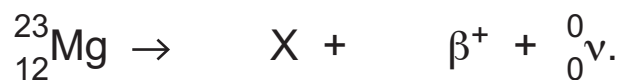
- (c)** A meson has a charge of $-1e$, where e is the elementary charge. The quark composition of the meson includes a charm antiquark.

State and explain a possible flavour (type) of the other quark in the meson.

[2]

23. O/N 23/P23/Q8/a,b

- 8 (a) The nuclide ${}^{23}_{12}\text{Mg}$ is an isotope of magnesium that undergoes β^+ decay to form a new nuclide X according to the equation



Four numbers are missing from the equation.

- (i) For the nuclide ${}^{23}_{12}\text{Mg}$, state what is represented by the numbers 23 and 12.

23 represents:

12 represents:

[2]

- (ii) Complete the equation by inserting the missing numbers.

[2]

- (iii) State the name of the group (class) of fundamental particles to which the positron and neutrino belong.

[1]

- (b) A radioactive source emits particles from its nuclei when it decays.

Fig. 8.1 shows, for the source, the variation with kinetic energy of the number of particles emitted.

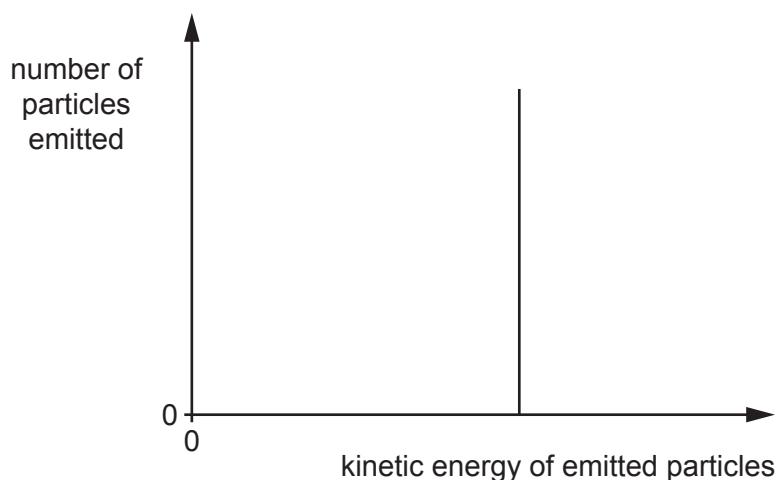


Fig. 8.1

State how Fig. 8.1 shows that these nuclei do **not** undergo beta-decay.

[1]

24. M/J 23/P22/Q8

booklet (scan + OCR)

- (a) Nucleus P and nucleus Q are isotopes of the same element.
Nucleus Q is unstable and emits a β^- particle to form nucleus R.

- (i) For nuclei P and Q, compare:
- the number of protons

.....

- the number of neutrons.

.....

[2]

- (ii) When nucleus Q decays to form nucleus R, the quark composition of a nucleon changes.
State the change to the quark composition of the nucleon.

..... [1]

- (iii) State the name of another particle that must be emitted from nucleus Q in addition to the β^- particle.

..... [1]

- (b) A hadron consists of two charm quarks and one bottom quark.
Determine, in terms of the elementary charge e , the charge of the hadron.

charge = e [2]**25. M/J 23/P21/Q8**

booklet (scan + OCR)

An isolated stationary nucleus X decays by emitting an α -particle to form a nucleus Y.
Nucleus Y and nucleus Z are isotopes of the same element.

- (a) By comparing the number of protons in each nucleus, state and explain whether the charge of nucleus Y is less than, greater than or the same as the charge of:

- (i) nucleus Z

..... [1]

- (ii) nucleus X.

.....

..... [2]

- (b) Use the principle of conservation of momentum to explain why nucleus Y cannot be stationary immediately after the decay of nucleus X.

.....

.....

26. M/J 23/P23/Q7

booklet (scan + OCR)

- (a) Table 7.1 shows incomplete data for three flavours (types) of quark. The elementary charge is e .

Table 7.1

flavour	quark		antiquark	
	symbol	charge/ e	symbol	charge/ e
up	u	$+\frac{2}{3}$	$\bar{u}$	
down	d		$\bar{d}$	
charm	c		$\bar{c}$	

Complete Table 7.1 by inserting the missing charges.

- (b) Using the symbols given in Table 7.1, state a possible quark combination for the following hadrons: [2]

(i) a neutral baryon

..... [1]

(ii) a meson with a charge of $+e$.

..... [1]

- (c) Quarks are fundamental particles.

Electrons are in another group (class) of fundamental particle.

(i) State the name of this group.

..... [1]

(ii) State the name of another particle in this group.

..... [1]

27. MAR 23/P22/Q7

booklet (scan + OCR)

Incomplete in the scanned booklet: its printed pages 534–535 are missing

- (a) Nuclei X and Y are different isotopes of the same element.

Nucleus X is unstable and emits a β^+ particle to form nucleus Z.

By comparing the number of protons in each nucleus, state and explain whether the charge of nucleus X is less than, the same as or greater than the charge of:

(i) nucleus Y

..... [1]

(ii) nucleus Z.

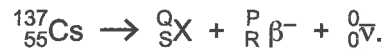
.....

..... [2]

28. M/J 22/P22/Q7

booklet (scan + OCR)

- (a) A nucleus of caesium-137 ($^{137}_{55}\text{Cs}$) decays by emitting a β^- particle to produce a nucleus of an element X and an antineutrino. The decay is represented by



- (i) State the number represented by each of the following letters.

P

Q

R

S [2]

- (ii) State the name of the class (group) of particles that includes the β^- particle and the antineutrino.

..... [1]

- (b) A particle Y has a quark composition of ddd where d represents a down quark.

A particle Z has a quark composition of $\bar{u}d$ where $\bar{u}$ represents an up antiquark.

- (i) Show that the charges of particles Y and Z are equal.

[2]

- (ii) State and explain which particle is a meson and which particle is a baryon.

meson:

baryon:

[2]

29. M/J 22/P21/Q7

booklet (scan + OCR)

(a) An unstable nucleus ${}^A_Z\text{X}$ decays by emitting a β^- particle.**(i)** Determine quantitatively the changes, if any, in A and Z when X decays.change in A =change in Z = [2]**(ii)** In addition to the β^- particle, another lepton is emitted during the decay.

State the name of the other lepton that is emitted.

..... [1]

(b) A particle P is composed of an up quark (u) and a down antiquark ($\bar{d}$).**(i)** Calculate the charge q of particle P in terms of e , where e is the elementary charge.

Show your working.

 $q = \dots\dots\dots e$ [2]**(ii)** Particle P belongs to **two** classes (groups) of particles.

State the names of these two classes.

1

2

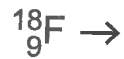
[2]

30. M/J 22/P23/Q7

booklet (scan + OCR)

- (a)** Fluorine-18 ($^{18}_9\text{F}$) is an isotope that decays to an isotope of oxygen (O) by the emission of a β^+ particle.

- (i)** Complete the nuclear equation for the decay, including all the particles involved.



- (ii)** A quark in the fluorine-18 nucleus changes flavour during the decay. State this change of flavour. [3]

..... quark to quark.

[1]

- (b)** A hadron has a charge of $-2e$, where e is the elementary charge.

- (i)** State and explain whether the hadron is a meson or a baryon.

.....

..... [2]

- (ii)** State a possible quark composition for the hadron.

..... [1]

31. MAR 22/P22/Q7

booklet (scan + OCR)

- (a) A nucleus of sodium-22 ($^{22}_{11}\text{Na}$) decays by emitting a β^+ particle. A different nucleus is formed by the decay.

(i) State the name of another lepton that is produced by the decay.

..... [1]

- (ii) Determine the nucleon number and the proton number of the nucleus that is formed by the decay.

nucleon number =

proton number = [2]

- (iii) The quark composition of a nucleon in the sodium-22 nucleus is changed during the decay.

Describe the change to the quark composition of the nucleon.

..... [1]

- (b) A baryon consists of quarks that are the same flavour (type). The charge of the baryon is $-2e$, where e is the elementary charge.

(i) Calculate, in terms of e , the charge of each quark.

charge = e [1]

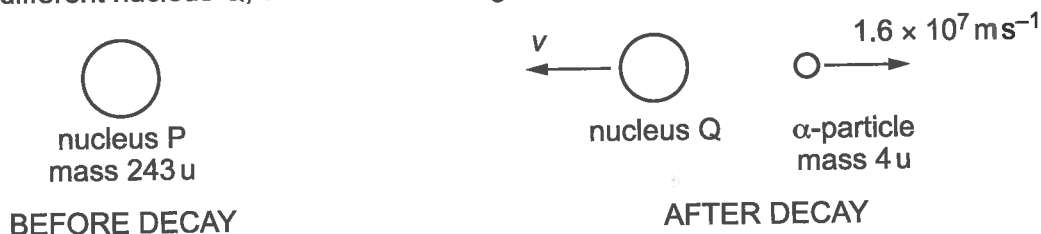
- (ii) State a possible flavour (type) of the quarks.

..... [1]

32. O/N 21/P22/Q7

booklet (scan + OCR)

A stationary nucleus P of mass $243u$ decays by emitting an α -particle of mass $4u$ to form a different nucleus Q, as illustrated in Fig. 7.1.

**Fig. 7.1**

The initial speed of the α -particle is $1.6 \times 10^7 \text{ ms}^{-1}$.

- (a) Use the principle of conservation of momentum to explain why the initial velocities of nucleus Q and the α -particle must be in opposite directions.

..... [2]

(b) Determine the initial speed v of nucleus Q.

$$v = \dots\dots\dots \text{ms}^{-1} \quad [2]$$

(c) Calculate the initial kinetic energy, in MeV, of the α -particle.

$$\text{kinetic energy} = \dots\dots\dots \text{MeV} \quad [3]$$

(d) A graph of number of neutrons N against proton number Z is shown in Fig. 7.2.

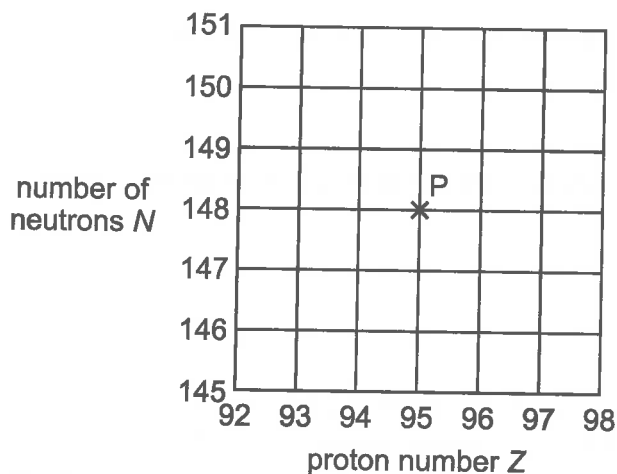


Fig. 7.2

The graph shows a cross that represents nucleus P.

A nucleus R has a nucleon number of 242 and is an isotope of nucleus P.

Nucleus R decays by emitting a β^- particle to form a different nucleus S.

(i) On Fig. 7.2, draw a cross to represent:

1. nucleus R (label this cross R)

2. nucleus S (label this cross S).

[2]

(ii) State the name of the other lepton, in addition to the β^- particle, that is emitted during the decay of nucleus R.

..... [1]

33. O/N 21/P21/Q6

booklet (scan + OCR)

- (a) Complete Table 6.1 to show the masses (in terms of the unified atomic mass unit u) and charges (in terms of the elementary charge e) of α , β^+ and β^- particles.

Table 6.1

	mass/ u	charge/ e
α -particle		
β^+ particle		
β^- particle		

[4]

- (b) Carbon-14 is radioactive and decays by emission of β^- particles.

- (i) Nuclei do not contain β^- particles.

Explain the origin of the β^- particle that is emitted from the nucleus during β^- decay.

..... [1]

- (ii) State the change in the quark composition of a carbon-14 nucleus when it emits a β^- particle.

..... [1]

- (iii) Suggest why the β^- particles are emitted with a range of different energies.

.....
 [2]

34. M/J 21/P22/Q6

booklet (scan + OCR)

May be outside the 2025–27 syllabus

- (a) One of the results of the α -particle scattering experiment is that a very small minority of the α -particles are scattered through angles greater than 90° .
State what may be inferred about the structure of the atom from this result.

.....
..... [2]

- (b) An α -particle is made up of other particles. One of these particles is a proton.
State and explain whether a proton is a fundamental particle.

..... [1]

- (c) A radioactive source produces a beam of α -particles in a vacuum. The average current produced by the beam is $6.9 \times 10^{-9} \text{ A}$.
Calculate the average number of α -particles passing a fixed point in the beam in a time of 1.0 minute.

number = [3]

- (d) The α -particles in the vacuum in (c) enter a uniform electric field. The α -particles enter the field with their velocity in the same direction as the field.
State and explain whether the magnitude of the acceleration of an α -particle due to the field decreases, increases or stays constant as the α -particle moves through the field.

.....
..... [2]

- (e) A nucleus X is an isotope of a nucleus Y. The mass of nucleus X is greater than that of Y. Both of the nuclei are in the same uniform electric field.
State and explain whether the magnitude of the electric force acting on nucleus X is greater than, less than or the same as that acting on nucleus Y.

.....
..... [2]

35. M/J 21/P21/Q6

booklet (scan + OCR)

May be outside the 2025–27 syllabus

(a) A proton in a nucleus decays to form a neutron and a β^+ particle.

(i) State the name of another lepton that is produced in the decay.

..... [1]

(ii) State the name of the interaction (force) that gives rise to this decay.

..... [1]

(iii) State which of the three particles (proton, neutron or β^+ particle) has the largest ratio of charge to mass.

..... [1]

(iv) Use the quark model to show that the charge on the proton is $+e$, where e is the elementary charge.

(v) The quark composition of the proton is changed during the decay. Describe the change to the quark composition. [2]

..... [1]

(b) A nucleus X ($^{12}_6\text{X}$) and a nucleus Y ($^{16}_8\text{Y}$) are accelerated by the same uniform electric field.

(i) Determine the ratio

$$\frac{\text{electric force acting on nucleus X}}{\text{electric force acting on nucleus Y}}$$

ratio = [2]

(ii) Determine the ratio

$$\frac{\text{acceleration of nucleus X due to the field}}{\text{acceleration of nucleus Y due to the field}}$$

ratio = [1]

(iii) Nucleus X is at rest in the uniform electric field at time $t = 0$.

The field causes nucleus X to accelerate so that it moves through the field.

On Fig. 6.1, sketch the variation with time t of the acceleration a of nucleus X due to the field.

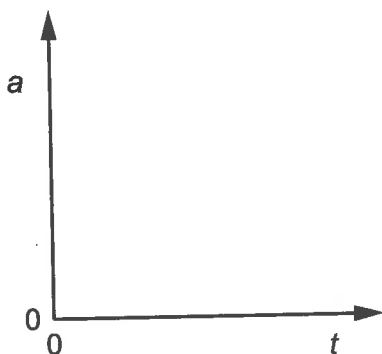


Fig. 6.1

[1]

36. M/J 21/P23/Q6

booklet (scan + OCR)

May be outside the 2025–27 syllabus

(a) State the quark composition of:**(i) a proton**

..... [1]

(ii) a neutron

..... [1]

(iii) an alpha-particle.

..... [2]

(b) In the alpha-particle scattering experiment, alpha-particles were directed at a thin gold foil.

State what may be inferred from:

(i) the observation that most alpha-particles pass through the foil

..... [1]

(ii) the observation that some alpha-particles are scattered through angles greater than 90°.

..... [2]

(c) A proton and an alpha-particle are moving in the same uniform electric field.

Determine the ratio

$$\frac{\text{acceleration of proton due to the electric field}}{\text{acceleration of alpha-particle due to the electric field}}$$

ratio = [2]

37. MAR 21/P22/Q7

booklet (scan + OCR)

- (a)** The results of the α -particle scattering experiment provide evidence for the structure of the atom.

Result 1: The vast majority of the α -particles pass straight through the metal foil or are deviated by small angles.

Result 2: A very small minority of α -particles is scattered through angles greater than 90° .

State what may be inferred (deduced) from:

- (i)** result 1

..... [1]

- (ii)** result 2.

..... [2]

- (b)** A radioactive decay sequence contains four nuclei, P, Q, R and S, as shown.



Nucleus S is an isotope of nucleus P.

- (i)** Determine the proton number and the nucleon number of nucleus S.

proton number =

nucleon number = [2]

- (ii)** The quark composition of a nucleon in Q changes as Q decays to form R.
Describe this change to the quark composition of the nucleon.

..... [1]

38. O/N 20/P22/Q7/a

booklet (scan + OCR)

- (a)** State a similarity and a difference between an up quark and an up antiquark.

similarity:

difference: [2]

39. O/N 20/P21/Q8/a,b

booklet (scan + OCR)

(a) State a similarity and a difference between a down quark and a down antiquark.

similarity:

difference: [2]

(b) For a nucleus of aluminium-25 (${}^{25}_{13}\text{Al}$):**(i)** state the number of protons and the number of neutrons

number of protons =

number of neutrons = [1]

(ii) show that the charge is $2.1 \times 10^{-18} \text{ C}$.

..... [1]

40. O/N 20/P23/Q7/c

booklet (scan + OCR)

(a) The nucleus is carbon-14 (${}^{14}_6\text{C}$). This nucleus decays to form a new nucleus by releasing a β^- particle and only one other particle of negligible mass.**(i)** Calculate the nucleon number and the proton number of the **new** nucleus.

nucleon number =

proton number = [1]

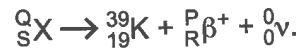
(ii) State the name of the particle of negligible mass.

..... [1]

41. M/J 20/P22/Q7

booklet (scan + OCR)

- (a) A nucleus of an element X decays by emitting a β^+ particle to produce a nucleus of potassium-39 (${}^{39}_{19}\text{K}$) and a neutrino. The decay is represented by



- (i) State the number represented by each of the following letters.

P

Q

R

S [2]

- (ii) State the name of the interaction (force) that gives rise to β^+ decay.

..... [1]

- (b) A hadron is composed of three identical quarks and has a charge of $+2e$, where e is the elementary charge.

Determine a possible type (flavour) of the quarks.
Explain your working.

.....

..... [2]

42. M/J 20/P21/Q6/b

booklet (scan + OCR)

May be outside the 2025–27 syllabus

(a) An uncharged atom of uranium-238 ($^{238}_{92}\text{U}$) has a change made to its number of orbital electrons. This causes the atom to change into a new particle (ion) X that has an overall charge of $+2e$, where e is the elementary charge.

(i) Determine the number of protons, neutrons and electrons in the particle (ion) X.

number of protons =

number of neutrons =

number of electrons = [3]

(ii) The particle (ion) X is in the electric field in **(a)** at a point midway between the plates. Determine the magnitude of the electric force acting on X.

force = N [2]

(iii) The nucleus of uranium-238 ($^{238}_{92}\text{U}$) decays in stages, by emitting α -particles and β^- particles, to form a nucleus of thorium-230 ($^{230}_{90}\text{Th}$).

Calculate the total number of α -particles and the total number of β^- particles that are emitted during the decay of uranium-238 to thorium-230.

number of α -particles =

number of β^- particles = [2]

43. M/J 20/P23/Q7/b,c

booklet (scan + OCR)

- (a) The β^- particle is emitted from the source with a kinetic energy of $3.4 \times 10^{-16} \text{ J}$.
Calculate the speed at which the β^- particle is emitted.

speed = ms^{-1} [2]

- (b) The β^- particle is produced by the decay of a neutron.

- (i) Complete the equation below to represent the decay of the neutron.



- (ii) State the name of the group (class) of particles that includes:

1. neutrons

.....

2. β^- particles.

..... [2]

44. O/N 19/P22/Q7

booklet (scan + OCR)

A nucleus of plutonium-238 (${}_{94}^{238}\text{Pu}$) decays by emitting an α -particle to produce a new nucleus X and 5.6 MeV of energy. The decay is represented by



- (a) Determine the number of protons and the number of neutrons in nucleus X.

number of protons =

number of neutrons = [2]

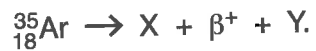
- (b) Calculate the number of plutonium-238 nuclei that must decay in a time of 1.0 s to produce a power of 0.15 W.

number = [2]

45. O/N 19/P21/Q7

booklet (scan + OCR)

(a) The decay of a nucleus ${}_{18}^{35}\text{Ar}$ by β^+ emission is represented by



A nucleus X and two particles, β^+ and Y, are produced by the decay.

State:

(i) the proton number and the nucleon number of nucleus X

proton number =

nucleon number =

[1]

(ii) the name of the particle represented by the symbol Y.

..... [1]

(b) A hadron consists of two down quarks and one strange quark.

Determine, in terms of the elementary charge e , the charge of this hadron.

charge = [2]

46. O/N 19/P23/Q7

booklet (scan + OCR)

A stationary nucleus of a radioactive isotope X decays by emitting an α -particle to produce a nucleus of neptunium-237 and 5.5 MeV of energy. The decay is represented by



- (a) Calculate the number of protons and the number of neutrons in a nucleus of X.

number of protons =

number of neutrons = [2]

- (b) Explain why the energy transferred to the α -particle as kinetic energy is less than the 5.5 MeV of energy released in the decay process.

..... [1]

- (c) A sample of X is used to produce a beam of α -particles in a vacuum. The number of α -particles passing a fixed point in the beam in a time of 30 s is 6.9×10^{11} .

- (i) Calculate the average current produced by the beam of α -particles.

current = A [2]

- (ii) Determine the total power, in W, that is produced by the decay of 6.9×10^{11} nuclei of X in a time of 30 s.

power = W [2]

47. M/J 19/P22/Q6

booklet (scan + OCR)

May be outside the 2025–27 syllabus

(a) State what is meant by a *field line* (*line of force*) in an electric field.

..... [1]

(b) An electric field has two different regions X and Y. The field strength in X is less than that in Y. Describe a difference between the pattern of field lines (lines of force) in X and in Y.

..... [1]

(c) A particle P has a mass of $0.15u$ and a charge of $-1e$, where e is the elementary charge.

(i) Particle P and an α -particle are in the same uniform electric field. Calculate the ratio

$$\frac{\text{magnitude of acceleration of particle P}}{\text{magnitude of acceleration of } \alpha\text{-particle}}.$$

ratio = [3]

(ii) Particle P is a hadron composed of only two quarks. One of them is a down (d) quark. By considering charge, determine a possible type (flavour) of the other quark. Explain your working.

.....

.....

..... [3]

48. M/J 19/P21/Q7

booklet (scan + OCR)

The booklet also files this question under Unit 2

- (a) One of the results of the α -particle scattering experiment is that a very small minority of the α -particles are scattered through angles greater than 90° .
State what may be inferred about the structure of the atom from this result.

.....
..... [2]

- (b) A hadron has an overall charge of $+e$, where e is the elementary charge. The hadron contains three quarks. One of the quarks is a strange (s) quark.

- (i) State the charge, in terms of e , of the strange (s) quark.

charge = [1]

- (ii) The other two quarks in the hadron have the same charge as each other.
By considering charge, determine a possible type (flavour) of the other two quarks.
Explain your working.

.....
..... [2]

49. M/J 19/P23/Q7

booklet (scan + OCR)

A sample of a radioactive substance may decay by the emission of either α -radiation or β -radiation and/or γ -radiation.

State the type of radiation, one in each case, that:

- (a) consists of leptons

..... [1]

- (b) contains quarks

..... [1]

- (c) cannot be deflected by an electric field

..... [1]

- (d) has a continuous range of energies, rather than discrete values of energy.

..... [1]

50. MAR 19/P22/Q7

booklet (scan + OCR)

(a) The names of four particles are listed below.

alpha beta-plus neutron proton

State the name(s) of the particle(s) in this list that:

(i) are not fundamental

..... [1]

(ii) do not experience an electric force when situated in an electric field

..... [1]

(iii) has the largest ratio of charge to mass.

..... [1]

(b) A hadron has a charge of $+e$ where e is the elementary charge. The hadron is composed of only two quarks. One of these quarks is an antidown ($\bar{d}$) quark. By considering charge, state and explain the name (flavour) of the other quark.

.....

..... [3]

51. O/N 18/P22/Q8

booklet (scan + OCR)

(a) In the following list, underline all particles that are leptons.

antineutrino positron proton quark

[1]

(b) A stationary nucleus of magnesium-27, ${}^{27}_{12}\text{Mg}$, decays by emitting a β^- particle and γ radiation.

An incomplete equation to represent this decay is

**(i)** State the nucleon number and the proton number of nucleus X.

nucleon number =

proton number = [2]

(ii) State the name of the interaction that gives rise to this decay.

..... [1]

(iii) State **two** possible reasons why the sum of the kinetic energy of the β^- particle and the energy of the γ radiation is less than the total energy released during the decay of the magnesium nucleus.

1.

2. [2]

52. M/J 18/P22/Q7

booklet (scan + OCR)

A stationary nucleus X decays to form nucleus Y, as shown by the equation

$$X \rightarrow Y + \beta^- + \bar{\nu}.$$

(a) In the above equation, draw a circle around all symbols that represent a lepton. [1]

(b) State the name of the particle represented by the symbol $\bar{\nu}$.

..... [1]

(c) Energy is released during the decay process. State the form of the energy that is gained by nucleus Y. [1]

.....

(d) By comparing the compositions of X and Y, state and explain whether they are isotopes.

.....

..... [2]
(e) The quark composition of one nucleon in X is changed during the emission of a β^- particle. Describe this change to the quark composition.

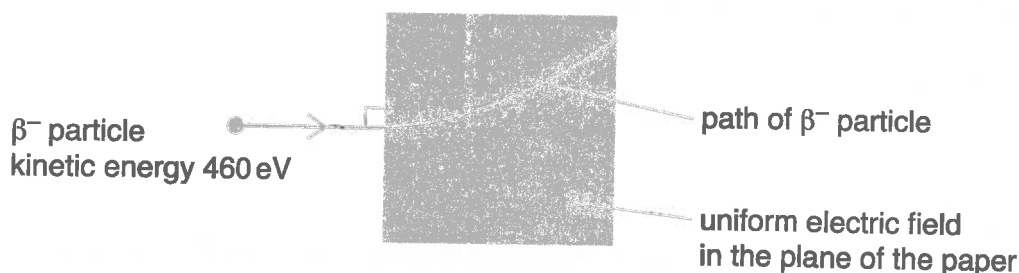
..... [1]

53. M/J 18/P21/Q7

booklet (scan + OCR)

May be outside the 2025–27 syllabus

A β^- particle from a radioactive source is travelling in a vacuum with kinetic energy 460 eV. The particle enters a uniform electric field at a right-angle and follows the path shown in Fig. 7.1.

**Fig. 7.1**

- (a) The direction of the electric field is in the plane of the paper.

On Fig. 7.1, draw an arrow to show the direction of the electric field.

[1]

- (b) Calculate the speed of the β^- particle before it enters the electric field.

speed = ms^{-1} [3]

- (c) Other β^- particles from the same radioactive source travel outside the electric field along the same incident path as that shown in Fig. 7.1.

State and briefly explain whether those β^- particles will all follow the same path inside the electric field.

.....

..... [2]

Answers Section

1. M/J 26/P22/Q6/a,c,d

6(a)	Most of the α -particles pass <u>straight</u> through (the atom) OR Most of the α -particles have (very) small deviations	B1	Allow 'majority' / 'nearly all' / 'large proportion' for 'most' Allow 'partcles' / 'them' to mean α -particles (BOD) Allow 'no deviation' / 'undeviated' / 'deviate by small / insignificant / minimal angles' instead of small deviations Allow 'deflection' instead of deviation Allow 'gold / leaf' for atom Ignore 'some' instead of 'most' Ignore 'a large <u>number</u> ' as this is not comparative Ignore just 'pass through' for 'pass straight through' Ignore 'only a (very) small proportion of alpha particles had (very) large deviations' as this does not necessarily mean the rest had little or no deviation
6(c)	3	A1	Allow '3 up' or '3u'
6(d)	One quark and one antiquark	B1	Allow 'quark and antiquark' (assume 1 of each) Allow $q\bar{q}$ BOD Ignore named quarks whether correct or incorrect e.g. 'up quark and antidown quark' as general composition required Ignore '2 quarks'

2. M/J 26/P23/Q5/a,b,c

5(a)	(a hadron consisting of) one quark and one antiquark	B1	Allow 'quark and antiquark' (assume 1 of each) Allow $q\bar{q}$ BOD Ignore named quarks whether correct or incorrect e.g. 'up quark and antidown quark' as general composition required. Ignore '2 quarks'
5(b)	quark charges have magnitudes of either $\frac{2}{3}(e)$ or $\frac{1}{3}(e)$	C1	May be implied by the use of these magnitude of charge (and only these charges) in the reasoning below.
	$-1e$: <i>either</i> : $-\frac{1}{3}(e)$, $-\frac{1}{3}(e)$ and $-\frac{1}{3}(e)$ or: $-\frac{2}{3}(e)$, $-\frac{2}{3}(e)$ and $+\frac{1}{3}(e)$	A1	Allow in terms of specific quarks provided magnitudes of charge are stated e.g. d has charge $-1/3(e)$ so ddd. Not e^- for e
	$+2e$: $+\frac{2}{3}(e)$, $+\frac{2}{3}(e)$ and $+\frac{2}{3}(e)$	A1	Not e^- for e
5(c)(i)	$P = 18$ and $R = 0$	A1	
	$Q = 7$ and $S = (+)1$	A1	
5(c)(ii)	(electron) neutrino	B1	
5(c)(iii)	up to down	B1	Allow u to d Allow uud to udd ignore any charges
5(c)(iv)	total energy emitted (in each decay) is constant	B1	
	one of : - particle Z takes some of the energy - the energy (emitted) is shared between Z and the β^+ particle - the way the total energy is distributed is different for each decay	B1	Ignore momentum arguments unless linked to KE Allow 'neutrino' for Z

3. M/J 26/P24/Q7/a,b

7(a)(i)	(different forms of the same) element / atom / nuclei with the same proton number	B1	Allow atomic number Ignore particle Ignore reference to electron number
	and different nucleon number / number of neutrons	B1	Allow mass number Allow number to be included once only

7(a)(ii)	minimum number of α particles = 1	A1	
	minimum number of β^- particles = 2	A1	
7(a)(iii)	$\begin{array}{c} 228 \\ \text{Th} \\ 90 \end{array}$	A1	No ecf from 7(a)(ii)
		A1	
7(b)(i)	up, up, down	B1	Allow u, u, d Allow any combination of words and symbols Allow arrows Ignore charges
7(b)(ii)	an up quark changes to a down quark	B1	Allow $u \rightarrow d$ Allow $uud \rightarrow udd$ Allow it changes to udd if 7b(i) is correct Ignore neutrino / antineutrino / beta-plus / beta-minus Allow arrows.

4. MAR 26/P22/Q7/a,b,c

7(a)	a <u>nucleus</u> containing <u>protons</u> and <u>neutrons</u>	B1
	(a nucleus) surrounded by (orbital) electrons	B1
7(b)(i)	alpha (decay)	B1
7(b)(ii)	beta-minus (decay)	B1
7(c)	ratio = $n \times \text{elementary charge} / \text{mass}$ (mass =) $(n \times) 1.60 \times 10^{-19} / 1.76 \times 10^{11} = 9.09 \times 10^{-31}$	B1
	OR (mass =) $(n \times) 1.60 \times 10^{-19} / (1.76 \times 10^{11} \times 1.66 \times 10^{-27}) = 0.000548$	(B1)
	OR (charge =) $1.76 \times 10^{11} \times 9.11 \times 10^{-31} = 1.60 \times 10^{-19}$	(B1)
	OR (ratio =) $1.60 \times 10^{-19} / 9.11 \times 10^{-31} = 1.76 \times 10^{11}$	(B1)
	(so X is a) positron	B1

5. O/N 25/P21/Q6/a,b,c

6(a)(i)	numbers of protons and electrons both = 1	A1
	number of neutrons = 2	A1
6(a)(ii)	diagram shows 2 neutrons and 1 proton labelled and forming a nucleus	B1
	diagram shows 1 electron labelled and separated from the nucleus (not touching the proton and neutrons)	B1
6(b)(i)	helium nucleus: top line = 3 and bottom line = 2	A1
	beta particle: top line = 0 and bottom line = -1	A1
6(b)(ii)	(electron) antineutrino	B1
6(c)	proton: up up down or neutron: up down down	C1
	(tritium:) 4 up, 5 down	A1

6. O/N 25/P22/Q6/a,b,c

6(a)	(a particle that) cannot be divided / subdivided (into smaller particles)	B1
------	---	-----------

6(b)(i)	any combination of one quark and one antiquark	C1
	up / charm / top and antiup / anticharm / antitop or down / strange / bottom and antidown / antistrange / antibottom	A1
6(b)(ii)	$E_{(K)} = \frac{1}{2} mv^2$	C1
	$2.1 \times 10^{-16} = \frac{1}{2} \times 0.67 \times 1.66 \times 10^{-27} \times v^2$	C1
	$v = 6.1 \times 10^5 \text{ m s}^{-1}$	A1
6(c)(i)	number of electrons = 88	A1
6(c)(ii)	(number of nucleons = $228 - 5 \times 4 = 208$) number of protons = $88 - (5 \times 2) + 4$ = 82	A1
	number of neutrons = $208 - 82 = 126$ = 126	A1

7. O/N 25/P24/Q6/a,b,c

6(a)(i)	W deflects upwards	B1
	W has deflection that is the mirror image of X	B1
	Y deflects downwards less than X or path of Y is parallel to particle Z	B1
6(a)(ii)	nucleus has most of the mass of the atom or nucleus is charged	B1
6(a)(iii)	atom is mostly empty space or nucleus occupies a <u>very</u> small proportion of the space in the atom	B1
6(b)	mass = 4 u	A1
	charge = (+)2e	A1
6(c)(i)	baryon	B1
6(c)(ii)	charge on up quark = $(+)\frac{2}{3}(e)$ and charge on down quark = $(-)\frac{1}{3}(e)$	C1
	(proton is up up down and neutron is up down down) $\left(2 \times \frac{2}{3}\right)e - \left(1 \times \frac{1}{3}\right)e = 1e$ and $\left(1 \times \frac{2}{3}\right)e - \left(2 \times \frac{1}{3}\right)e = 0$	A1

8. M/J 25/P21/Q7/a,b

7(a)	${}^{12}_7\text{N} \rightarrow {}^{12}_6\text{C} + {}^0_1\text{B}^+ + {}^0_0\text{n}$	B1
	beta-plus shown	
	neutrino shown	B1
	symbols, nucleon numbers and proton numbers all correct	B1
7(b)(i)	+1	B1
7(b)(ii)	Lepton(s)	B1

7(b)(iii)	flavour	charge / e	
	up / u	$-\frac{2}{3}$	
	up / u	$-\frac{2}{3}$	
	down / d	$(+)\frac{1}{3}$	
	3 correct quark flavours		B1
	Charge on anti-up quark $-\frac{2}{3}(e)$		B1
	Charge on anti-down quark $(+)\frac{1}{3}(e)$		B1

9. M/J 25/P22/Q7/a,b,c

7(a)	(a particle that) cannot be divided/subdivided (into smaller particles)	B1
7(b)(i)	$(p \times e) / (14u) = 4.1 \times 10^7$	C1
	$p = (4.1 \times 10^7 \times 14 \times 1.66 \times 10^{-27}) / (1.60 \times 10^{-19})$	C1
	$p = 6$ (answer should be an integer)	A1
7(b)(ii)	${}^{14}_6\text{X} \rightarrow {}^{14}_7\text{Z}$	B1
	${}^0_{-1}\text{e}^{(-)} \text{ or } {}^0_{-1}\beta^{(-)}$	B1
	${}^0_0\bar{\nu}_{(e)}$	B1
7(c)	(the nuclei are undergoing) β^+ decay	B1
	A correct explanation in terms of charge and a correct explanation in terms of energy Explanations in terms of charge: • (particles / decay) positively charged so cannot be β^- • (particles / decay) positively charged so could be / is β^+ • β^- (particles / decay) are negatively charged • β^+ (particles / decay) are positively charged Explanations in terms of energy: • range of energies so not α (particles / decay) • range of energies so is β (particles / decay) • β (particles / decay) have a range of energies • α (particles / decay) have discrete energies / not range of energies	B1

10. M/J 25/P23/Q8/a,b

8(a)	flavour	charge / e	
	up / u	$-\frac{2}{3}$	
	down / d	$(+)\frac{1}{3}$	
	down / d	$(+)\frac{1}{3}$	
	3 correct quark flavours		B1
8(b)	particle: (electron) neutrino		B1
	antiparticle: positron		B1

11. M/J 25/P24/Q7/a,b,c

7(a)	(electron) neutrino (electron) antineutrino - electron - positron Any two, 1 mark each	B2
7(b)	only meson <u>and</u> neutron are underlined	B1
7(c)	(quark composition is) up, up, down / uud $(\text{charge}) = \frac{2}{3}e + \frac{2}{3}e - \frac{1}{3}e = (+1)e = (+) 1.6 \times 10^{-19} \text{ (C)}$	B1
	or $(\text{charge}) = 1.07 \times 10^{-19} + 1.07 \times 10^{-19} - 5.33 \times 10^{-20} = (+) 1.6 \times 10^{-19} \text{ (C)}$	A1

12. MAR 25/P22/Q7/a,c

7(a)	nucleon number of Q = 226 <u>and</u> proton number of R = 86	B1
7(c)	Any three from: - momentum - charge - nucleon number - neutron number - proton number	B3

13. O/N 24/P21/Q5/a,b

5(a)(i)	$P = 40$ and $R = 0$ $Q = 20$ and $S = -1$	A1
5(a)(ii)	(electron) antineutrino	B1
5(a)(iii)	leptons	B1
5(b)	(alpha particle consists of) 2 protons and 2 neutrons	C1
	quark composition of proton = up up down or quark composition of neutron = up down down	C1
	quark composition (of alpha particle) = 6 up, 6 down	A1

14. O/N 24/P22/Q6/a,b

6(a)	mass of α (particle) is <u>much</u> greater (than β^+ particle) both particles are positively charged (magnitude of) charge on α (particle) is twice the charge (on β^+ particle)	B1
6(b)(i)	cross labelled Q at (82, 212)	B1
6(b)(ii)	particles emitted are: - beta-minus (particle) / electron (electron) antineutrino either particle named	B1
	both particles named and no incorrect particles named	B1

15. O/N 24/P23/Q7/a,b,c

7(a)	up quark charge = $(+) \frac{2}{3}$ and down quark charge = $-1/3$	B1
	strange quark charge = $-1/3$	B1
	up antiquark charge = $-2/3$ and down antiquark charge = $(+) \frac{1}{3}$ and strange antiquark charge = $(+) \frac{1}{3}$	B1
7(b)(i)	hadron(s)	B1
7(b)(ii)	baryons composed of three quarks or baryons composed of three antiquarks	B1
	mesons composed of one quark and one antiquark	B1
7(c)	up quark changes to a down quark	B1
	positron and (electron) neutrino (emitted)	B1

16. M/J 24/P21/Q7/a,b

7(a)	${}^{66}_{29}\text{Cu}$	B1
7(b)(i)	the energy of the decay is fixed / constant	B1
	the energies of the beta particles have a (continuous) range of values / varies / not constant	B1
	another particle / an (anti)neutrino must possess the extra / remaining energy (difference between energy of the decay and the β kinetic energy)	B1
7(b)(ii)	(electron) antineutrino	B1
7(b)(iii)	${}^{66}_{29}\text{Cu} \rightarrow {}^{66}_{30}\text{Zn} + {}^0_{-1}\beta + {}^0_{0}\bar{\nu}_e$	B1
	values for Cu and Zn correct with no other extra particles on either side of the equation	
	second term correct (${}^0_{-1}\beta$)	B1
	third term correct (${}^0_{0}\bar{\nu}_e$)	B1

17. M/J 24/P22/Q7/a,b

7(a)	meson(s) or hadron(s)	B1
7(b)	Q: anti-up or anti-charm or anti-top	B1
	R: up or charm or top	B1

18. M/J 24/P23/Q6/a,b,c

6(a)	${}^{147}_{62}\text{Sm}$	A1
6(b)(i)	the (kinetic) energy of the particles is discrete / has only one value (so must be alpha)	B1
	and beta particles have a (continuous) range of (kinetic) energies (so can't be beta)	B1
6(b)(ii)	${}^{147}_{62}\text{Sm} \rightarrow {}^{143}_{60}\text{Nd} + {}^4_2\alpha$	A1
	values for Sm and Nd correct with no other extra particles on either side of the equation	
	${}^4_2\alpha$ correct	A1

6(c)(i)	up quark charge is $+(2/3)(e)$ or bottom quark charge is $-(1/3)(e)$	C1
	$0 = +(2/3)(e) - (1/3)(e) + q$	A1
	(so) charge (on third quark must be) $-(1/3)(e)$	
6(c)(ii)	down or strange	A1

19. MAR 24/P22/Q4/a

4(a)	${}^4_2\alpha$	B1
	${}^{211}_{82}\text{Q}$	B1

20. MAR 24/P22/Q8/a,b,c

8(a)	lepton(s)	B1
8(b)(i)	up or top or charm	B1
8(b)(ii)	meson(s)	B1
8(c)(i)	β^- (particle) or electron	B1
8(c)(ii)	equal	B1
8(c)(iii)	(the charge of) R is greater (than Q)	B1

21. O/N 23/P21/Q7/a,b,c,d

7(a)	a (very) small proportion of (alpha) particles are deflected (back) through large angles / angles greater than 90°	B1
	a large proportion of (alpha) particles pass straight through / deflected by small angles	B1
7(b)	electromagnetic wave / electromagnetic radiation	A1
7(c)	neutrino classified as a lepton only <u>and</u> positron classified as a lepton only	B1
	neutron classified as a baryon and a hadron and not as a lepton	B1
7(d)(i)	1.32 u	A1
7(d)(ii)	working states or implies 3 quarks and each quark has a charge of $(+)\frac{2}{3}(e)$	B1
	any combination of 3 quarks comprised of one or more of up / charm / top	B1

22. O/N 23/P22/Q7/a,b,c

7(a)	only baryon and nucleon underlined	B1
7(b)	number of α -particles = 6	B1
	number of β^- particles = 4	B1
7(c)	charm antiquark (charge) = $-\frac{2}{3}(e)$	B1
	or	
	$-\frac{2}{3}(e) + q = -1(e)$	
	(other quark $q = -\frac{1}{3}(e)$ so) down / strange / bottom	B1

23. O/N 23/P23/Q8/a,b

8(a)(i)	23: nucleon number or number of neutrons and protons	B1
	12: proton number or number of protons	B1

8(a)(ii)	both numbers correct for X: ${}_{11}^{23}\text{X}$	B1
	both numbers correct for beta-plus: ${}^0_1\beta^+$	B1
8(a)(iii)	lepton(s)	A1
8(b)	(the emitted particles) have a single (kinetic) energy or beta particles / decay have a (continuous) range of (kinetic) energies	B1

24. M/J 23/P22/Q8

- (a) (i) number of protons: equal/same
number of neutrons: unequal/different [2]
- (ii) down (quark) changes to up (quark)
or
up down down (quarks) change to up up down (quarks) [1]
[1]
- (iii) (electron) antineutrino
- (b) charm (quark charge) is $(+)2/3(e)$
or
2 charm (quark charges) is $(+)4/3(e)$
or
bottom (quark charge) is $-1/3(e)$
charge = $+2/3(e) + 2/3(e) - 1/3(e)$
= $(+)1(e)$ [2]

25. M/J 23/P21/Q8

- (a) (i) Y and Z have equal numbers of protons and (so) they have the same charge [1]
- (ii) Y has (two) fewer protons (than X)
(so) Y has less charge (than X) [2]
- (b) (total) momentum before decay is zero
or
X has zero / no momentum
(total momentum after decay must be zero so)
Y must have equal (and opposite) momentum to α -particle
(so cannot be stationary / must have speed/velocity) [2]

26. M/J 23/P23/Q7

- (a) down charge = $-1/3(e)$ and charm charge = $(+)2/3(e)$
all antiquarks have opposite sign and same (non-zero) magnitude of charge as the
corresponding quarks [2]
- (b) (i) udd or cdd [1]
- (ii) $u\bar{d}$ or $c\bar{d}$ [1]
- (c) (i) lepton(s) [1]
- (ii) positron / neutrino / antineutrino [1]

27. MAR 23/P22/Q7

- (a) (i) X has same number of protons as Y (and so) charge of X is the same as the charge of Y [1]
- (ii) X has (one) more proton (than Z)
(so) X has greater charge (than Z) [2]
- (b) (i) meson(s) [1]
- (ii) one quark and one antiquark [1]

28. M/J 22/P22/Q7

- (a) (i) $P = 0$ and $Q = 137$
 $R = -1$ and $S = 56$ [2]
- (ii) lepton(s) [1]
- (b) (i) (charge of ddd / Y) $= -\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e) = -1(e)$
(charge of $\bar{u}d$ / Z) $= -\frac{1}{3}(e) - \frac{2}{3}(e) = -1(e)$ [2]
- (ii) meson: Z / $\bar{u}d$ because consists of a quark and an antiquark [2]
baryon: Y / ddd because consists of three quarks [1]

29. M/J 22/P21/Q7

- (a) (i) change in $A = 0$
change in $Z = (+)1$ [2]
- (ii) (electron) antineutrino [1]
- (b) (i) up / u (charge) $= (+)\frac{2}{3}e$ or antidown / $\bar{d} = (+)\frac{1}{3}e$
or
 $(q) = \frac{2}{3}e + \frac{1}{3}e$ [2]
 $q = (+)1e$
- (ii) hadron(s) [2]
meson(s)

30. M/J 22/P23/Q7

- (a) (i) ${}^{18}_{9}\text{F} \rightarrow {}^{18}_{8}\text{O} + {}^{0}_{(+1)}\beta^{(+)} + {}^{(0)}_{(0)}\nu_{(e)}$
 ν or neutrino (B1)
 ${}^{0}_{(+1)}\beta^{(+)} (B1)$ [3]
 ${}^{18}_{8}\text{O} (B1)$ [1]
- (ii) up quark to down quark
- (b) (i) must be three (anti)quarks as largest (negative) quark charge is $(-)\frac{2}{3}(e)$
or
mesons can only have a charge of 0 or $\pm 1(e)$
(so hadron is) a baryon [2]
- (ii) any combination of three from:
antiup (quark) / up antiquark
and/or anticharm (quark) / charm antiquark [1]
and/or antitop (quark) / top antiquark

31. MAR 22/P22/Q7

- (a) (i) (electron) neutrino [1]
 (ii) nucleon number = 22 [2]
 proton number = 10
 (iii) up up down changes to up down down
 or [1]
 up changes to down [1]
- (b) (i) charge = $-\frac{2}{3}e$ [1]
 (ii) antiup / anticharm / antitop [1]

32. O/N 21/P22/Q7

- (a) (total) momentum before (decay) is zero
 or
 P has zero momentum
 (total momentum after decay must be zero so)
 α -particle and Q have momenta in opposite directions
 (and therefore velocities are in opposite directions) [2]
- (b) $p = 239 \text{ (u)} \times v$ or $4 \text{ (u)} \times 1.6 \times 10^7$
 $239 \text{ (u)} \times v = 4 \text{ (u)} \times 1.6 \times 10^7$
 $v = 2.7 \times 10^6 \text{ m s}^{-1}$ [2]
- (c) $E_{(K)} = \frac{1}{2}mv^2$
 $= \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times (1.6 \times 10^7)^2$
 $= 8.5 \times 10^{-13} \text{ (J)}$
 $= 8.5 \times 10^{-13} / 1.60 \times 10^{-13} \text{ (MeV)}$
 $= 5.3 \text{ MeV}$
- (d) (i) 1. R plotted at (95,147) [3]
 2. S plotted at (96,146)
 (ii) (electron) antineutrino [2]
 [1]

33. O/N 21/P21/Q6

- (a) α -particle mass given as 4u
 α -particle charge given as (+)2e
 both β -particles mass given as 0.0005u
 β^+ charge given as (+)e and β^- charge given as $-e$
 (Completed table:

	mass / u	charge / e
α	4	(+)2
β^+	0.0005	(+)1
β^-	0.0005	-1

- (b) (i) neutron decays into proton and an electron / β^- particle [4]
 (ii) down to up [1]
 (iii) (electron) antineutrino(s) emitted [1]
 energy (released in decay)/momentum shared between antineutrino and β^- particle [2]

34. M/J 21/P22/Q6

- (a) the nucleus is charged
the majority of the mass (of atom) is in the nucleus [2]
- (b) made up of quarks (so) not a fundamental particle [1]
- (c) ($Q =$) $6.9 \times 10^{-9} \times 60$
number = $(6.9 \times 10^{-9} \times 60) / (2 \times 1.60 \times 10^{-19})$
= 1.3×10^{12} [3]
- (d) (magnitude of electric) force is constant [3]
(so magnitude of) acceleration is constant [2]
- (e) (nuclei have) same charge/same number of protons
(so) same (magnitude of) force [2]

35. M/J 21/P21/Q6

- (a) (i) (electron) neutrino [1]
(ii) weak (nuclear force/interaction) [1]
(iii) β^+ (particle) [1]
(iv) (quark structure is) up up down or uud
 $(2/3)e + (2/3)e - (1/3)e = (+)e$ [2]
(v) up up down changes to up down down or $uud \rightarrow udd$
or
up changes to down or $u \rightarrow d$ [1]
- (b) (i) $F = Eq$
ratio = $6/8$
= 0.75 [2]
(ii) ratio = $0.75 \times (16/12)$
= 1.0 [1]
(iii) horizontal straight line at a non-zero value of a [1]

36. M/J 21/P23/Q6

- (a) (i) up up down [1]
(ii) up down down [1]
(iii) (alpha-particle is) 2 protons and 2 neutrons
6 up, 6 down [2]
- (b) (i) most of an atom is empty space
or
the nucleus (volume) is (very) small compared with the atom [1]
(ii) the nucleus is charged
the majority of the mass of atom is in the nucleus [2]
- (c) $F = Eq$ and $a = F/m$
 $a = Eq/m$
ratio = $(e/m) / (2e/4m)$
= 2 [2]

37. MAR 21/P22/Q7

- (a) (i) most of the atom is empty space
or
the nucleus (volume) is very small compared to the atom [1]
- (ii) the nucleus is charged
the mass is concentrated in nucleus / small region / small volume / small core
or
the majority of the mass is in nucleus / small region / small volume / small core [2]
- (b) (i) proton number = 84 [2]
nucleon number = 214
- (ii) up down down changes to up up down / udd $\rightarrow$ uud
or
down changes to up / d $\rightarrow$ u [1]

38. O/N 20/P22/Q7/a

- (a) similarity: same/equal mass or
same/equal (magnitude of) charge or
both fundamental (particles)
difference: opposite (sign of) charge
or
one is matter and the other is antimatter [2]

39. O/N 20/P21/Q8/a,b

- (a) similarity: same/equal mass
or
same/equal (magnitude of) charge
or
both fundamental (particles)
difference: opposite (sign of) charge
or
one is matter and the other is antimatter [2]
- (b) (i) number of protons = 13 and number of neutrons = 12 [1]
(ii) (charge =) $13 \times 1.60 \times 10^{-19} \text{ (C)} = 2.1 \times 10^{-18} \text{ (C)}$ [1]

40. O/N 20/P23/Q7/c

- (a) (i) nucleon number = 14
and
proton number = 7 [1]
- (ii) (electron) antineutrino [1]

41. M/J 20/P22/Q7

- (a) (i) P = 0 and Q = 39
R = (+)1 and S = 20 [2]
- (ii) weak (nuclear force/interaction) [1]
- (b) charge of quark(s) = (+) $2e / 3$
up/u (quarks) [2]

42. M/J 20/P21/Q6/b

- (a) (i) number of protons = 92
 number of neutrons = 146
 number of electrons = 90 [3]
- (ii) $F = EQ$
 $= 1.5 \times 10^4 \times 2 \times 1.60 \times 10^{-19}$
 $= 4.8 \times 10^{-15} \text{ N}$ [2]
- (iii) number of α -particles = 2
 number of β^- particles = 2 [2]

43. M/J 20/P23/Q7/b,c

- (a) $E = \frac{1}{2}mv^2$
 $3.4 \times 10^{-16} = \frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$
 $v = 2.7 \times 10^7 \text{ ms}^{-1}$ [2]
- (b) (i) ${}^1_1\text{p}$
 ${}^{0-}_{0}\text{V(e)}$ [2]
- (ii) 1. hadrons [2]
 2. leptons

44. O/N 19/P22/Q7

- (a) number of protons = 92
 number of neutrons = 142 [2]
- (b) $5.6 \text{ MeV} = 5.6 \times 1.60 \times 10^{-19} \times 10^6 (= 8.96 \times 10^{-13} \text{ J})$
 number $= 0.15 / (5.6 \times 1.60 \times 10^{-13})$
 $= 1.7 \times 10^{11}$
 or
 $0.15 \text{ W} = 0.15 / (1.60 \times 10^{-19} \times 10^6) (= 9.38 \times 10^{11} \text{ MeV s}^{-1})$
 number $= 9.38 \times 10^{11} / 5.6$
 $= 1.7 \times 10^{11}$ [2]

45. O/N 19/P21/Q7

- (a) (i) proton number = 17
 and
 nucleon number = 35 [1]
- (ii) (electron) neutrino [1]
- (b) d/down (quark charge) is $-\frac{1}{3}(e)$
 or
two d/down (quark charges) is $-\frac{2}{3}(e)$
 or
 s/strange (quark charge) is $-\frac{1}{3}(e)$
 charge $= -\frac{1}{3}(e) - \frac{1}{3}(e) - \frac{1}{3}(e)$
 $= -1(e)$ [2]

46. O/N 19/P23/Q7

- (a) number of protons = 95 [2]
 number of neutrons = 146

- (b) Np/neptunium (nucleus) has kinetic energy
 or [1]
 gamma/ γ -radiation produced

- (c) (i) $I = NQ / t$
 $I = (6.9 \times 10^{11} \times 2 \times 1.60 \times 10^{-19}) / 30$ [2]
 $= 7.4 \times 10^{-9} \text{ A}$
- (ii) $P = (6.9 \times 10^{11} \times 5.5 \times 10^6 \times 1.60 \times 10^{-19}) / 30$ [2]
 $= 0.020 \text{ W}$

path/direction in which a (free) positive charge will move [1]

47. M/J 19/P22/Q6

- (a) path/direction in which a (free) positive charge will move [1]
 (b) (lines) closer together in Y/further apart in X

- (c) (i) $a = Eq / m$
 or
 $F = Eq$ and $F = ma$
 ratio $= (1e / 0.15 \text{ u}) \times (4 \text{ u} / 2e)$ or $1 / 0.15 \times 4 / 2$
 ratio $= 13$

- (ii) down quark charge is $-(1/3)e$ [3]
 $-(1/3)e + q = -1e$ so $q = -(2/3)e$
 $-(2/3)e$ is anti-up / $\bar{u}$ (quark) (allow charm or top antiquark) [3]

48. M/J 19/P21/Q7

- (a) nucleus is charged
 the mass is concentrated in (very small) nucleus
 or
 the majority of the mass is in (very small) nucleus [2]
- (b) (i) $-(1/3)e$ [2]
 (ii) $2q - (1/3)e = e$ so $q = (2/3)e$ [1]
 up/u (quarks) (allow charm or top quarks) [2]

49. M/J 19/P23/Q7

- (a) beta/ β [1]
 (b) alpha/ α
 (c) gamma/ γ [1]
 (d) beta/ β [1]
 [1]

50. MAR 19/P22/Q7

- (a) (i) alpha, neutron and proton [1]
 (ii) neutron [1]
 (iii) beta plus or β^+ [1]
 (b) $\bar{d}$ has charge $(+)\frac{1}{3}e$
 (so) other quark has charge $= e - \frac{1}{3}e$
 $= (+)\frac{2}{3}e$
 other quark is an up / u [3]

51. O/N 18/P22/Q8

- (a) antineutrino **and** positron both underlined (and no other particles) [1]
 (b) (i) nucleon number = 27 [1]
 proton number = 13 [2]
 (ii) weak (nuclear force/interaction) [1]
 (iii) an (electron) antineutrino / $\bar{\nu}_{(e)}$ is produced (and this has energy) [1]
 X has kinetic energy [2]

52. M/J 18/P22/Q7

- (a) circle(s) drawn only around β^- and $\bar{\nu}$ symbols [1]
 (b) (electron) antineutrino [1]
 (c) kinetic (energy) [1]
 (d) Y has one more proton (and one less neutron)/X has one less proton (and one more neutron) [1]
 or
 Y has more protons (and fewer neutrons)/X has fewer protons (and more neutrons)
 or
 a neutron changes to a proton
 or
 the number of protons increases [2]
 (so) not isotopes
 (e) up down down changes to up up down or $udd \rightarrow uud$ [1]
 or
 down changes to up or $d \rightarrow u$

53. M/J 18/P21/Q7

(a) arrow pointing vertically down the page

[1]

(b) $E = \frac{1}{2}mv^2$

$$E = 460 \times 1.60 \times 10^{-19} \quad (= 7.36 \times 10^{-17} \text{ (J)})$$

$$v = [(2 \times 460 \times 1.60 \times 10^{-19}) / (9.11 \times 10^{-31})]^{1/2}$$

$$= 1.3 \times 10^7 \text{ ms}^{-1}$$

[3]

(c) β^- particles have range of/different/various speeds/velocities/momenta/energies
so they follow different paths

[2]